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Seminal Fluid Mediates Sexual Conflict Over Ejaculate Fate

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Abstract:	In species with multiple mating, males and females often differ in their reproductive interests, and so the transfer, storage and use of sperm can be a source of considerable sexual conflict [1,2], driving cycles of sexually antagonistic co-evolution [3–6]. Seminal fluid is likely to be a key mediator of such conflicts, since it is uniquely positioned to influence female reproductive physiology and behaviour after mating [7–11]. However, owing to the complexity, rapid evolution and functional redundancy of seminal fluid [10,12,13], identifying which seminal fluid proteins induce specific physiological or behavioural responses in mating partners is challenging. We therefore employed a quantitative genetics approach in combination with targeted gene knockdown in the simultaneously hermaphroditic flatworm Macrostomum lignano to first identify and then functionally characterise candidate seminal fluid transcripts inhibiting one striking post-mating behaviour. This so-called 'suck' behaviour occurs when worms respond to ejaculate receipt by placing their pharynx over the female genital opening, seemingly attempting to remove sperm and/or other ejaculate components [5,14,15]. Based on sexual conflict theory, we would expect selection for the ability of sperm donors to manipulate recipients to prevent this behaviour. In line with this, we found negative genetic correlations between the expression levels of six (out of 58) seminal fluid transcripts and the partner's suck propensity. Knockdown using RNA interference confirmed that two of these transcripts, which we here designate suckless-1 and suckless-2, indeed caused mating partners to suck less often. We have thus identified a male counter-adaptation to recipient suck behaviour, which itself likely evolved as a female counter-adaptation in the ongoing evolutionary conflict to (re)gain control over ejaculate fate.
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The Editors,
Current Biology

Bielefeld, 28 February 2019

Seminal Fluid Mediates Sexual Conflict Over Ejaculate Fate

Dear Editors,

Following our presubmission enquiry last week and Dr. Maderspacher's positive response, we are now submitting our manuscript to be considered for publication as a Report in Current Biology.

In this study, we provide experimental evidence for a novel biochemical adaptation of seminal fluid to manipulate the behaviour of mating partners. We think this would be of interest to the broad readership of Current Biology for two main reasons:

First, we employ what we believe is a novel and potentially widely applicable methodology to identify functional candidate genes, based on genetic correlations estimated between a gene expression network and the target functional trait.

Second, the function of male seminal fluid proteins we have thereby identified represents a striking effect on the mating partner's reproductive behaviour. This has broad implications for our understanding of male-female reproductive interactions and fertility, and specifically advances the field of seminal fluid studies by revealing how seminal fluid functions evolve through sexual conflict.

As previously communicated, we think potentially appropriate reviewers with expertise in seminal fluid and/or sexual conflict would include:

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Please note that we have no directly related manuscripts in press or under consideration elsewhere, but three unpublished studies from our lab are referred to in the manuscript: one (Ramm et al.) will appear shortly in Molecular Ecology and is a transcriptomics study that includes reference to the plastic expression of seminal fluid transcripts; a second (Weber et al.) is being revised for Evolution and describes an RNAi screen for seminal fluid-mediated fitness effects in *M. lignano*; and a third (Patlar & Ramm) is a manuscript currently in preparation that is a follow-up to the present study and investigates natural variation in the expression of the two *suckless* transcripts. We have not at present provided any of these in the submission, but would be happy to do so if you think this would be useful.

Many thanks for considering our manuscript and we look forward to hearing from you in due course.

Yours sincerely,

Steven Ramm

Seminal Fluid Mediates Sexual Conflict Over Ejaculate Fate

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SUMMARY

In species with multiple mating, males and females often differ in their reproductive interests, and so the transfer, storage and use of sperm can be a source of considerable sexual conflict [1,2], driving cycles of sexually antagonistic co-evolution [3–6]. Seminal fluid is likely to be a key mediator of such conflicts, since it is uniquely positioned to influence female reproductive physiology and behaviour after mating [7–11]. However, owing to the complexity, rapid evolution and functional redundancy of seminal fluid [10,12,13], identifying which seminal fluid proteins induce specific physiological or behavioural responses in mating partners is challenging. We therefore employed a quantitative genetics approach in combination with targeted gene knockdown in the simultaneously hermaphroditic flatworm *Macrostomum lignano* to first identify and then functionally characterise candidate seminal fluid transcripts inhibiting one striking post-mating behaviour. This so-called 'suck' behaviour occurs when worms respond to ejaculate receipt by placing their pharynx over the female genital opening, seemingly attempting to remove sperm and/or other ejaculate components [5,14,15]. Based on sexual conflict theory, we would expect selection for the ability of sperm donors to manipulate recipients to prevent this behaviour. In line with this, we found negative genetic correlations between the expression levels of six (out of 58) seminal fluid transcripts and the partner's suck propensity. Knockdown using RNA interference confirmed that two of these transcripts, which we here designate *suckless-1* and *suckless-2*, indeed caused mating partners to suck less often. We have thus identified a male counter-adaptation to recipient suck behaviour, which itself likely evolved as a female counter-adaptation in the ongoing evolutionary conflict to (re)gain control over ejaculate fate.

KEYWORDS

evolution, mating behaviour, reproduction, seminal fluid, sexual conflict, sexually antagonistic coevolution, simultaneous hermaphroditism, sperm competition

RESULTS AND DISCUSSION

Although sexually antagonistic adaptations in seminal fluid have been primarily investigated in species with separate sexes (e.g. [3,8,16,17]), inter-locus sexual conflict also occurs in hermaphrodites [7,18–20]. The precise adaptive significance of the suck behaviour in *M. lignano* (see Video S1) has not yet been established, but it is hypothesised to have evolved as a response to sexual conflict over control of received ejaculates [14,15]. Such a function might generally be predicted in the context of cryptic female choice [21], but it may be especially likely in hermaphrodites. Here, the higher motivation to donate sperm rather than receive it and frequent evolutionary outcome of 'reciprocal mating' [7,22] – where both mating partners donate and receive sperm – means there should be strong selection on recipients to gain control over processing received ejaculate components, either because (excess) sperm receipt itself is costly [18,22,23] or because the ejaculate contains manipulative seminal fluid proteins [24,25]. This should in turn create strong selection for counter-adaptations in donors, potentially also mediated through seminal fluid.

Consistent with the large number of seminal fluid proteins (SFPs) in other taxa (e.g. [26–29]), we already know that seminal fluid in *M. lignano* is complex, with at least 76 prostate-limited transcripts recently identified [30]. A large-scale RNAi screen provided some evidence that loss of specific SFPs can affect reproductive success [31], but importantly only for a minority of proteins investigated and without addressing their functions. To overcome the general problem of narrowing down the number of seminal fluid candidates with the goal of screening for specific functions, we therefore here sought to develop a method to more efficiently target a relevant subset of candidate SFPs in *M. lignano* that could manipulate the suck behaviour.

In the absence of genome-wide sequence information, one promising approach based on quantitative genetics recently advocated by Ayroles et al. [32] is to use genetic variation in gene expression to annotate novel genes based on genetic correlations with genes whose functions have already been established. Here, we extended this approach for identifying candidate genes by instead directly estimating genetic correlations between gene expression and the phenotype of interest. More specifically, to identify genes that might function as donor counter-adaptations that seek to prevent the suck response, we employed a two-step genetic correlation analysis. First, we used a seminal fluid transcript expression network [33] to initially home in on major axes of seminal fluid variation associated with partner suck propensity and second we estimated genetic correlations between transcripts loaded on these axes and suck propensity in order to identify specific candidate transcripts. We then tested the functional role of these candidates using RNA interference gene knockdown of donors followed by a behavioural assay of partner suck propensity.

Quantitative genetics identifies candidate seminal fluid transcripts mediating sexual conflict over ejaculate fate

We used quantitative genetic data from a panel of 12 inbred *M. lignano* lines, for which we recently described sources of genetic and environmental variation in the expression of 58 seminal fluid transcripts (summarised by four principal components) [33]. Environmental effects were estimated as differential expression at two social group sizes (pairs and octets) differing in sperm competition intensity (e.g. [34–39]). We measured the suck propensity of the mating partner as the proportion of copulations that were followed by a suck by that recipient worm in a standardised behavioural assay of mating pairs, where the donor worm came from one of these inbred line/group size combinations. To justify our approach of evaluating genetic correlations between partner suck propensity and seminal fluid transcript expression in donors, we first tested for genetic variation in partner suck propensity using inbred line identity (i.e. genotype) as a fixed effect, and in addition group size and genotype x group size interaction effects. Social group size had no effect on suck propensity of the recipient (two-way ANOVA, $F_{1,254} = 0.01$, $P = 0.91$) nor was there a genotype-by-group size interaction ($F_{11,254} = 1.6$, $P = 0.11$), but we found a highly significant genotype effect ($F_{11,254} = 3.2$, $P < 0.001$), indicating that recipients differed markedly in their suck propensity depending on the donor's genotype.

Having established genetic variation in both seminal fluid transcript expression [33] and partner suck propensity, we proceeded to identifying candidate transcripts based on genetic correlations. First, we calculated the genetic correlation between partner suck propensity and four major axes of seminal fluid transcript expression variation reported in Patlar et al. [33]. We did this in each of the two social group sizes (based on mean values of the traits per inbred line and group size level). This revealed evidence for a negative genetic correlation between the first principal component (PC1) of variation in seminal fluid transcript expression and partner suck propensity in both environments, with overlapping 95% confidence intervals (pairs: -0.72, 0.07; octets: -0.81, -0.23) suggesting this may be a consistent effect, but the correlation appeared on average stronger and significant in octets (octets: $r_G = -0.55$, $P < 0.01$) (Figures 1A,B). Given the higher intensity of sperm competition in octets [36], and thus presumably the greater premium on maintaining ejaculates in storage, this environment may be functionally more relevant to manipulating suck propensity, and we therefore prioritised this environment for subsequent analyses.

Since the majority of seminal fluid transcripts are significantly loaded on PC1, which thus describes overall seminal fluid transcript abundance [33], this indicates that partners of genotypes with higher investment in overall seminal fluid expression suck less often. We therefore next estimated the genetic correlations between suck propensity and expression level of each of 58 individual seminal fluid transcripts in octets. Genetic correlations coefficients were negative for the majority of the transcripts (ranging between -0.69 and 0.40, Table S1), but were especially strong – and significant after False-Discovery Rate (FDR) correction – for six transcripts: *Mlig-pro5*, *Mlig-pro6*, *Mlig-pro30*, *Mlig-pro31*, *Mlig-pro32* and *Mlig-pro33* (Figure 1C; see Weber et al. [30] for details regarding seminal fluid transcript nomenclature in *M. lignano*). We thus next tested whether targeted knockdown of any of these six candidate transcripts, individually or in combination, affected recipient suck propensity.

RNA interference confirms two seminal fluid transcripts mediate sexual conflict over ejaculate fate

To evaluate the hypothesised effect of the six candidate SFP transcripts on partner suck propensity in a manipulative experiment, we used RNAi to generate donor worms lacking these proteins in their ejaculates. To do so, we took advantage of the regenerative capacity of this flatworm [40,41] to amputate the posterior tail region of adults, thereby 'resetting' their seminal fluid production. We then allowed them to regenerate whilst soaking in i) dsRNA probes specific to one of the six candidate SFP transcripts (Table S2); ii) a seventh 'combined' treatment dsRNA from all six candidate SFP transcripts; or iii) a negative control treatment (i.e., in the absence of dsRNA). Following RNAi treatment, the knockdown and control donors were tested for their effects on partner suck propensity using a behavioural assay in which the (treated) donor worm was paired with a standardised (partner) recipient worm for 2.5 hrs in a 3.4 µl drop of artificial sea water squeezed between two glass slides, a well-established technique in *M. lignano* that enables the continuous monitoring and recording of mating behaviour [14]. We then scored numbers of copulations and the frequency with which they were followed by a partner suck during the whole 2.5 hrs period. We used this as the dependent variable to test whether any of the RNAi treatments differed from controls in the suck propensity of their partners.

As predicted, we found that two of the six candidates indeed resulted in a significantly higher suck propensity of partners, based on comparing each RNAi knockdown treatment to the control (Figure 2; independent Welch's two sample t-tests, followed by FDR correction, *Mlig-pro31*: $t_{54.84} = 2.60$, $P_{\text{adjusted}} = 0.03$; *Mlig-pro32*: $t_{46.55} = 2.92$, $P_{\text{adjusted}} = 0.02$; all other single knockdowns $P > 0.6$). Because worms mated to donors lacking *Mlig-pro31* and *Mlig-pro32* in their ejaculate are more likely to perform the suck behaviour, these results strongly imply that these two seminal fluid transcripts normally function to reduce partner suck propensity, and so we would now designate them as *suckless-1* and *suckless-2*, respectively. Both exhibit evidence for being secreted [30] and they are highly positively correlated in their expression in *M. lignano* [33]. However, these transcripts show no sequence similarity and when blasted against the *M. lignano* genome assembly ML2 [42], they align to two different genomic regions, which further suggests that they indeed belong to different genes.

The combined knockdown of all six candidates had a comparable effect on suck propensity to the single knockdown of *suckless-1* and *suckless-2* ($t_{50.96} = 3.18$, $P = 0.02$). This could indicate that these transcripts do not impact additively on recipient suck behaviour, although one caveat here is that we observed some residual signal of *suckless-2* expression in combined knockdown worms in the *in situ* hybridization screening we performed to verify RNAi efficiency (Figure S1). Note also that none of the RNAi treatments differed from controls for either total mating frequency (Table S3A) or the donor's own suck propensity (Table S3B), indicating that the effects we have identified act specifically on recipient suck propensity. These effects are also consistent with a previous report in *M. lignano* that showed worms suck less often after copulating with virgin partners than sexually experienced partners [43]. Virgin worms had a higher fill grade of their prostate glands and so presumably transferred more seminal fluid compared to sexually experienced worms [43].

Given the likely impact of suck behaviour on received ejaculates, we would expect that the ability to manipulate this behaviour could have direct consequences for the donor's competitive fertilisation success (paternity share) if it affects the total or relative number of donor sperm present at fertilization. As a first test of this, after the behavioural assay was completed, we paired each recipient worm with a standardised GFP-transformed competitor for 2 hours, using the GFP phenotype of offspring to assign paternity and thereby estimate 'defensive' sperm competitive ability of the donor (P_1) [44–46]. Somewhat surprisingly, we saw no marked effect of RNAi knockdown on paternity success (Table S4). This may have been due to our specific experimental design, and the fact that the high mating rate of *M. lignano* might have meant all donors maximised sperm in storage by the end of the extended 2.5h mating period, irrespective of their RNAi treatment and thus partner suck propensity. Moreover, a subsequent experiment investigating natural variation in *suckless-1* and *suckless-2* expression generated through genotype x environment interactions has revealed a strong positive correlation between *suckless-1* expression and P_1 [47], further supporting the functional significance of the behavioural manipulation we have identified.

Suck propensity is genetically correlated with ejaculate investment

In addition to our main hypothesis, we were also interested to explore the interrelationships between seminal fluid investment, suck propensity of the recipient and the other key aspect of male allocation, i.e., sperm production, as estimated by testis size of donors (e.g. [34–36,48]). For testis size, we observed significant genotype ($F_{11,257} = 9.0$, $P < 0.001$), and social group size ($F_{1,257} = 11.4$, $P = 0.001$) effects, but no interaction ($F_{11,257} = 1.1$, $P = 0.25$). We found average testis size increases with increasing group size (mean testis area in pairs = $14.2 \pm 0.6 \times 10^3$ µm^2, in octets = $16.8 \pm 0.7 \times 10^3$ µm^2) as expected from previous studies (e.g. [36,37,39,49]). Within octets, testis size was positively genetically correlated with overall seminal fluid investment (PC1; Figure 1A, $r_G = 0.56$, $P < 0.01$), which could explain why we also observed a negative correlation between testis size and partner suck propensity (Figure 1A, $r_G = -0.41$, $P = 0.02$). This means genotypes with larger testes (and thus probably more sperm) likely also transfer more manipulative seminal fluid proteins in competitive environments and serves to emphasize the integrated nature of the male ejaculate (e.g. [50–52]). If sperm and seminal fluid investment are positively correlated [52], this could also help explain a previous pattern observed in *M. lignano* whereby suck propensity depended on the sex allocation (i.e., the relative investment into sperm and egg production) of mating pairs [34].

CONCLUSIONS

Although seminal fluid is a potentially crucial mediator of reproductive interactions with important implications for fertility [53–56], post-mating sexual selection [57–61], sexually antagonistic coevolution [3,9,16,17] and ultimately even speciation [29,62,63], its rapid evolution [64–66] and highly complex nature [12,26–28,67] means only a small subset of seminal fluid components have been functionally well characterised in any experimental system. To overcome this difficulty of linking seminal fluid functions to specific proteins contained within it, we here demonstrate the utility of using quantitative genetics coupled with targeted gene knockdown, an empirical approach that could potentially be applicable to a wide range of taxa, including non-model organisms for which alternative methods are not feasible. Adopting such a strategy, we were able to identify two seminal fluid transcripts in *M. lignano*, namely *suckless-1* and *suckless-2*, that appear to mediate sexual conflict over ejaculate fate by manipulating the partner's post-mating behaviour and causing them to suck less often. Donors potentially thereby regain control over the retention of their own sperm or other ejaculate components and in so doing enhance their own reproductive success. This suggests that the evolution of these proteins was a further step in the ongoing cycles of sexually antagonistic coevolution driving seminal fluid diversification.

Experimental Procedures

Study organism

Macrostomum lignano (Rhabditophora, Platyhelminthes) is a simultaneous hermaphroditic, outcrossing and reciprocally copulating, free-living marine flatworm [68]. It is cultured in the laboratory in artificial sea water (32‰ salinity) at 20°C, 14:10 hrs light:dark cycle, under 60% humidity and fed with the diatom *Nitzschia curvilineata*. Worms copulate frequently under laboratory conditions (ca. ten matings per hour) and sperm is transferred together with seminal fluid via the stylet (male copulatory organ) to the partner's female antrum (sperm storage organ) [14,15]. Seminal fluid is produced by prostate gland cells located around the stylet [15,69], and includes a complex mixture of proteins [30]. The worms used in this experiment originated from 12 different inbred lines that belong to a larger set of inbred lines which was originally generated at the University of Innsbruck and is now maintained at the University of Basel [70]. These inbred lines are called DV lines, and we here used the lines DV1 [36,42], DV3, DV8, DV13, DV28, DV69 and DV71 [71] and DV47, DV49, DV61, DV75 and DV84 [70].

Social group size manipulation

In our previous study, we described the social group size manipulation in detail [33]. Briefly, we raised 3-4 days old juveniles from 12 inbred lines for 51 days either in groups of two worms or groups of eight worms (referred to here as pairs and octets, respectively). The juvenile worms were randomly distributed into the pair or octet treatment level with a fully crossed design where all inbred line/group size combinations were represented. At age 51 days old, the morphological and behavioural measurements reported here were carried out as described below before the same worms were prepared for the seminal fluid transcript expression measurements reported in the previous study [33].

Observation of suck behaviour

One individual (hereafter focal individual) was randomly chosen from 18 independent pair replicates and 18 independent octet replicates from each inbred line and paired with a standardized mating partner (recipient) to evaluate genetic variation in the suck propensity of partners. The standardized recipients originated from one inbred line (DV1). To ensure their virginity, recipients were grown under strictly isolated conditions (in individual wells of 24-well tissue culture plates [TPP AG, Trasadingen, Switzerland] filled with 1 ml of 32‰ ASW and ad libitum food). Before the pairing, we labelled recipients by using food colour to distinguish them (and their suck behaviour) from the focal donor [43]. For labelling, we transferred the virgin worms to fresh 60 well HLA Terasaki Plates (Greiner Bio-One, Frickenhausen, Germany) including 3 µl colour solution (5 mg of Colorant Alimentaire Grand Blue, Les Artistes, Paris, France per one ml 32‰ ASW) and 7 µl 32‰ ASW with ad libitum food for 24 hrs. Then worms were transferred to fresh 24-well plates without colour solution and food (including only 1 ml of 32‰ ASW) for a few minutes to wash away colour solution residues. The colouring has no effect on worms' survival, fecundity or mating behaviour [43].

For the suck behaviour observations of the recipients, each focal worm and its labelled partner was transferred into mating observation chambers described in detail elsewhere [14]. Briefly, the labelled partner was taken and transferred with a 1.7 µl drop of ASW to the centre of a coated (Sigmacote®, Saint Louis, USA) microscope slide which had four HERMA photo stickers (Filderstadt, Germany) adhered to the sides (two per side). Then immediately one focal was transferred with 1.7 µl ASW onto the top of the coloured partner, merging the two drops. After ca. 12 drops were placed on the same slide, we sealed around the drops with a thin line of Vaseline to protect drops from evaporation and added a second slide on top of the drops, forming each drop into a shallow three-dimensional pool that the worms could swim inside. Immediately after preparation of one observation chamber, it was placed under a camera for a video recording of 2 hrs for suck behaviour genetic correlation analyses or 2.5 hrs for the knockdown behaviour assays described below. The video recordings were done with a DFK 41AF02 Camera (Imaging Source GmbH, Bremen, Germany) connected to a computer running the Debut Video Capture Software Professional, version 2.02. The videos were captured at one frame per second, with a frame rate of ten (a video of 2 hrs was zipped into 12 mins) and saved in .mov format with 1920 x 1080 HD resolution. Videos were analysed by using Kinovea video player, version 0.8.15. Recordings were analysed by an experimenter blind with respect to the different group size treatments and inbred lines, to count both the number of copulations and the occurrence of subsequent suck behaviour of coloured recipients. Any mating pairs in which no copulation occurred were excluded from the dataset (ca. 35% of the data).

Testis size and seminal fluid expression measurements

The transparent body of *M. lignano* allows non-invasive measurements of internal reproductive organs [49]. Briefly, after the video recordings, focal worms were taken out of the mating chamber and anesthetized in a 5:3 mixture of 7.14% $MgCl_2$ and 32‰ ASW for 10 minutes. Thereafter, we squeezed the focal donor dorsoventrally to a standard thickness between a microscope slide and a cover slip and took pictures acquired under 400x magnification using a Nikon Ni-U microscope (Nikon GmbH, Düsseldorf, Germany) connected to a ProgRes MFcool camera (Jenoptik GmbH, Jena, Germany). Images were processed for the testis area measurements, representing testis size, using the free-draw tools of the software ImageJ [72], blind with respect to group sizes and identity of inbred lines.

Immediately after imaging, focal worms were transferred individually to 24-well plates including 500 µl 32‰ ASW for about five minutes to wash out $MgCl_2$ and allow worms to recover from this anaesthetic. Afterwards, individual worms were transferred to 1 ml tubes containing 25 µl of RNALater solution (Ambion, Austin, USA), which immediately stabilizes RNA needed for SFP transcript expression measurements as described in detail in our previous study [33]. Here, we reused the expression data of single SFP transcripts, as well as data on principal component analyses derived from these that were originally reported in Patlar et al. [33], to test our hypothesis that the expression of some genes could negatively correlate with partner suck propensity.

RNA interference knockdown treatment and behavioural assays

In total, 576 juveniles from a stock culture of the DV1 inbred line were collected at 3-4 days old, and randomly distributed into three glass Petri dishes with *ad libitum* food. Taking advantage of the regenerative capacity of *M. lignano* [40], at age 50 days, we amputated half of the adult worms (equally and randomly picked from each Petri dish) to remove received and stored sperm and seminal fluid contents from their previous matings, as well as eggs produced, by cutting away their tail plate just posterior to the end of the ovaries. Afterwards, we isolated these worms for 15 days to be used as standardized recipients (mating partners) of the knockdown focal donors (see next paragraph). Isolation was conducted in 10 µl 32‰ ASW with ad libitum food in 60 well Terasaki plates to let them regenerate and produce fresh eggs. Worms were regularly fed until the mating observations. These worms were labelled blue as described above on day 14.

We amputated the second half of the adult worms at age 55 days to prepare RNA interference knockdown treatment groups. RNAi was performed as previously described [73]. Briefly, a group of double-stranded RNA (dsRNA) probes was generated for the candidate transcripts using specific primer pairs (Table S2). The dsRNAs were diluted in 32‰ ASW including algae to a final concentration of 25 ng/µl. The second half of the amputated worms were treated with dsRNA in Terasaki plates, where we put one worm in each well and randomized RNAi treatment across wells and plates. The dsRNA treatment was done by transferring worms to fresh wells with fresh dsRNA solution every second day for ten days, sufficient to guarantee complete regeneration but knockdown of the target transcript(s) [40]. In this way, we treated

36 worms with dsRNA for each of six candidates, plus another 36 worms were treated with a combined dsRNA solution (a mixed dsRNA solution of all candidates, hereafter referred to as the combined treatment). We also included 36 worms as a negative control group treated at the same time, under the same conditions but with the solution including only 32‰ ASW with algae. Some worms (42 out of 288) died during this process, seemingly independent of RNAi treatment or control group.

We performed video recordings on surviving individuals after the regeneration period. On day 65, the video recordings were performed in exactly the same way as described above for control and knockdown worms as focals, paired randomly with one of the tail-amputated and regenerated labelled worms. Control and knockdown worms were randomly assigned into mating chambers and observations were done blind to RNAi and control treatment groups (15/246 pairs where mating did not occur were excluded). To verify the success of RNAi knockdown treatments, we performed *in situ* hybridization (ISH) screening for all RNAi knockdown treatments of candidate transcripts individually, as well as for the combined knockdown of all candidates (Figure S1). ISH screening was performed as described in a recent study of seminal fluid in *M. lignano* [30].

Defensive sperm competitive ability (P_1) assay

After 2.5 hrs video recording of mating pairs of focal and labelled worms, we transferred labelled recipients into single wells of Terasaki plates and let them mate with standardized competitors for a further 2 hrs. The half-an-hour difference between mating pairs of focal and competitor was preferred because there is on average high sperm displacement and thus second male precedence in *M. lignano* [74]. Therefore, we aimed to equalise sperm competitiveness of the first and second partners to some extent, since we were here only interested in differences between treatment groups as first mates. We used transgenic worms that carry a Green-Fluorescent Protein (GFP) marker as competitors, originating from the outbred transgenic BAS1 culture obtained from a stock maintained at the Zoological Institute at University of Basel, Switzerland [70,75]. The GFP allele is a dominant marker that is expressed in all somatic and gametic cell types and therefore allows for easily genotyping offspring and assigning paternity following double mating experiments of a wild type, GFP(-) focal worm (here the RNAi knockdown or control) and a GFP(+) competitor worm [42,74].

GFP(+) competitors were prepared in the same way as recipient worms, i.e. first amputated and then isolated until forming the mating pairs for ten days. We scored paternity by counting GFP(-) offspring sired by focal and GFP(+) offspring sired by competitors by following the first ten days of egg laying by recipients, until all laid eggs had eventually hatched. In addition, we isolated GFP(+) competitors after pairing them with recipients to assess the offspring number produced by them, to exclude data of where a competitor worm did not lay eggs (assuming therefore that mating did not occur unless recipient also did not produce GFP(+) offspring). In total, nine out of 233 competitors did not produce any offspring, and their recipients also did not produce GFP(+) offspring; these were excluded.

Statistical analyses

The effect of social group size and genotype on recipient suck propensity and testis size was assessed using two-way ANOVA with interaction after testing for homogeneity of variance for social group size using Levene tests. Variances were found to be similar for both suck propensity ($P = 0.50$) and testis size ($P = 0.16$).

We estimated genetic correlations (r_G) by calculation of Pearson product-moment correlations of the inbred line means and calculated standard errors from 10,000 replicates of bootstrapped coefficients. We tested for statistical significance by comparing the *z*-scores to two-tailed significance levels derived from a standard normal distribution.

We performed Welch two sample *t*-tests, which is a more reliable test when the variances and sample size are unequal across groups, to compare mean partner suck propensity (and of donor as supplemental data, as well as mating frequency) of the different RNAi knockdown groups to the control group. We calculated adjusted *P*-values by performing Benjamini-Hochberg False Discovery Rate corrections to decrease the chance of Type I errors while identifying candidate transcripts and where multiple tests were applied.

For the comparison of paternity success of RNAi knockdown groups with the control group, we used a generalized linear model with a quasi-binomial error distribution and logit link function (for handling proportional data including many zeroes and ones), in which the response is a matrix where the first column is the number of the focal donor's offspring and the second column is the number of GFP(+) offspring. All statistical analyses were performed with R version 3.4.2 [76].

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Author Contributions

BP and SAR conceived the study; BP, MW and TT conducted the experiments; BP analysed the data; BP and SAR drafted the manuscript.

Declaration of Interests

The authors declare no competing interests.

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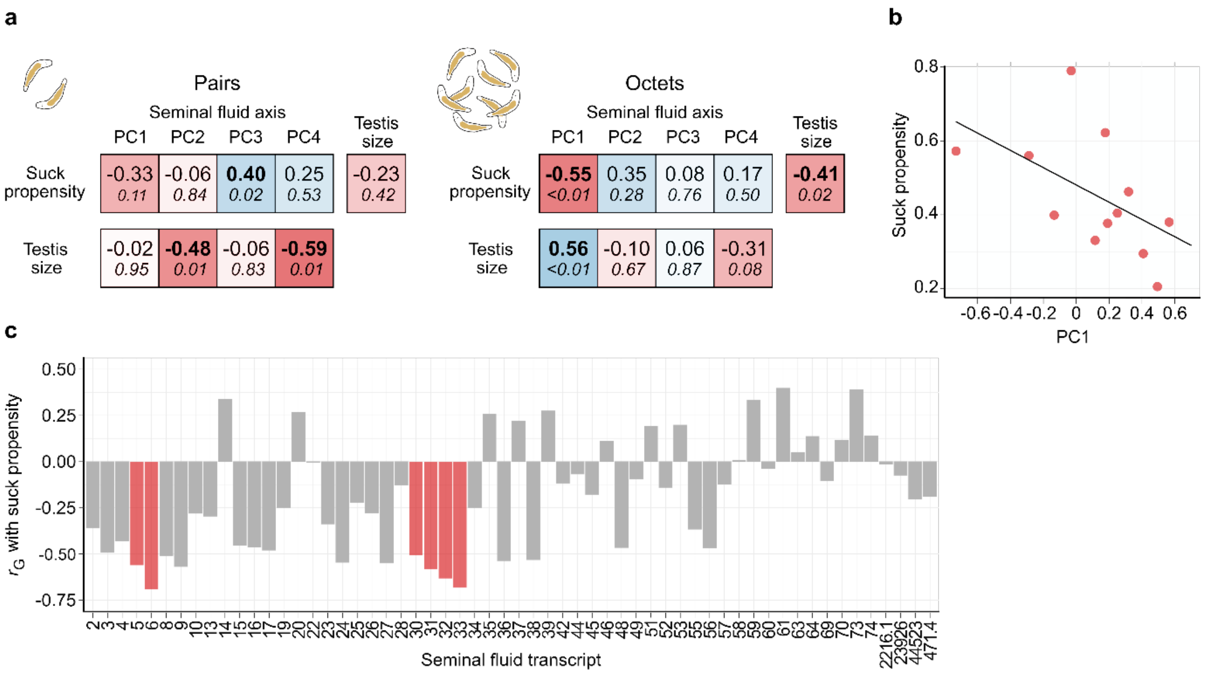

Figure 1: Genetic correlations (r_G) between seminal fluid expression of the donor and suck propensity of the recipient.

a) Genetic correlations (r_G) and *P*-values (italics) for main axes (principal components) of variation in seminal fluid transcript expression, testis size and suck propensity of mating partners in pairs and octets calculated as Pearson's product-moment correlation coefficients based on average genotype values at each group size. b) Scatterplot of the average suck propensity and PC1 scores of the 12 genotypes in octets (r_G = -0.55, P < 0.01). c) Genetic correlations calculated between expression of individual transcripts and suck propensity of recipients for 58 seminal fluid transcripts (Pearson's product-moment correlation coefficients based on average genotype values in octets). Red bars show significant correlations (candidate transcripts) derived from adjusted *P*-values after performing False Discovery Rate corrections. Numbers on the x-axis represent the identity of the different seminal fluid transcripts (*Mlig-pro1* – *Mlig-pro76*) from [30]. In addition, four prostate-limited transcripts (2216.1, 23926, 44523 and 471.4) are depicted in the figure (with their transcript IDs from the MLRNA110815 transcriptome assembly [52]) that were not initially identified as putative seminal fluid transcripts [33].

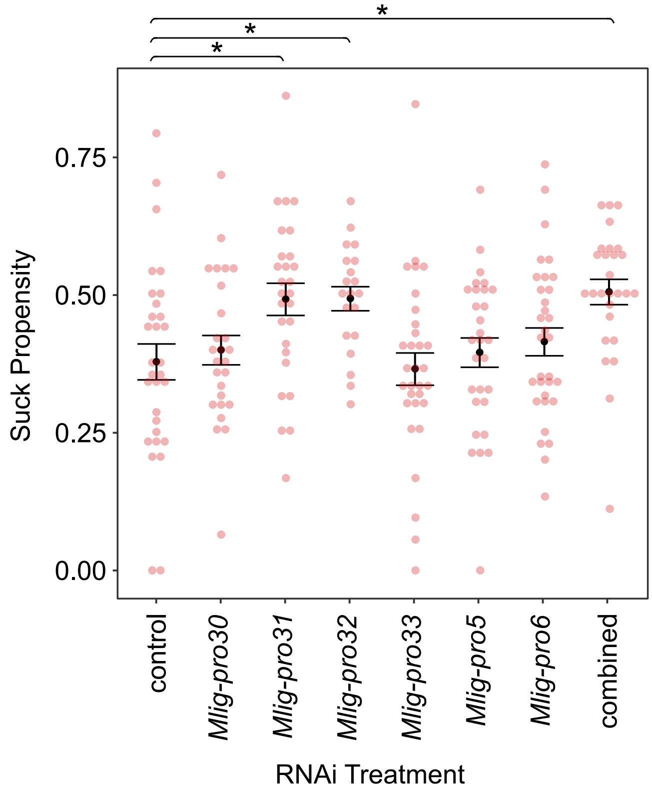

Figure 2: Identification of two key seminal fluid transcripts influencing partner suck behaviour following RNAi knockdown of candidates.

The figure shows suck propensity (the propensity of mating partners to perform the suck behaviour in 2.5 hrs) of each RNAi knockdown treatment and the treatment of combined knockdown of all candidate transcripts vs. the negative control group. Error bars represent standard errors of the mean suck propensity for each treatment (Black: control, grey: knockdown, light-grey: combined). Jittered red points are the underlying raw data, depicting the observed suck propensity of each individual sample.

Seminal Fluid Mediates Sexual Conflict Over Ejaculate Fate

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Supplemental Information

Video S1. Related to Figures 1 and 2: Suck behaviour in *Macrostomum lignano*. The video shows a representative single mating between two worms that is followed by the suck behaviour being performed by one of them. The worms begin mating after ca. 10s, separate at ca. 22s and immediately afterwards (ca. 23s) the worm to the upper-right of the video performs the suck behaviour (which lasts ca. 7s). For a more detailed description of the mating behaviour of *M. lignano*, see Schärer et al. (2004).

(.mov file supplied separately)

Table S1. Related to Figure 1: Genetic correlation coefficients (r_G) between donor-induced recipient suck propensity and donor expression levels of 58 seminal fluid transcripts when worms were kept in octets. *P*-values are derived from Pearson product moment correlation tests (alternative hypothesis: true correlation differs from 0), standard errors (se)were determined from distributions of 1000 bootstrap values. Adjusted *P*-values were calculated by performing False Discovery Rate analysis. Significant genetic correlations after FDR correction (5% cut-off) are displayed in bold. Transcripts 2216.1, 23926, 471.4 and 44523 are labelled with their transcript IDs from the MLRNA110815 transcriptome assembly [49].

SFP transcripts	r_G	*se*	*P*	Bootstrapped *P*	Adjusted *P*
Mlig-pro2	-0.36	0.3046	0.25	0.30	0.58
Mlig-pro3	-0.49	0.3017	0.10	0.03	0.13
Mlig-pro4	-0.43	0.3016	0.16	0.05	0.17
Mlig-pro5	**-0.56**	**0.3005**	**0.06**	**<0.001**	**0.003**
Mlig-pro6	**-0.69**	**0.2999**	**0.01**	**<0.001**	**<0.001**
Mlig-pro8	-0.51	0.3010	0.09	0.01	0.08
Mlig-pro9	-0.57	0.3001	0.05	0.01	0.05
Mlig-pro10	-0.28	0.3017	0.38	0.22	0.58
Mlig-pro13	-0.29	0.3029	0.35	0.29	0.58
Mlig-pro14	0.34	0.3002	0.28	0.23	0.58
Mlig-pro15	-0.45	0.2996	0.14	0.04	0.16
Mlig-pro16	-0.46	0.3013	0.13	0.05	0.17
Mlig-pro17	-0.48	0.3029	0.12	0.06	0.20
Mlig-pro19	-0.25	0.2992	0.43	0.32	0.60
Mlig-pro20	0.27	0.3006	0.40	0.51	0.78
Mlig-pro22	-0.01	0.3029	0.99	0.99	0.99
Mlig-pro23	-0.34	0.3003	0.28	0.30	0.58
Mlig-pro24	-0.55	0.3016	0.07	0.04	0.17
Mlig-pro25	-0.22	0.2994	0.49	0.53	0.78
Mlig-pro26	-0.28	0.3010	0.38	0.28	0.58
Mlig-pro27	-0.55	0.3025	0.06	0.02	0.11
Mlig-pro28	-0.13	0.3037	0.70	0.57	0.80
Mlig-pro30	**-0.51**	**0.2996**	**0.09**	**0.002**	**0.02**
Mlig-pro31	**-0.58**	**0.3013**	**0.05**	**<0.001**	**0.004**
Mlig-pro32	**-0.63**	**0.3018**	**0.03**	**<0.001**	**<0.001**
Mlig-pro33	**-0.68**	**0.2991**	**0.01**	**<0.001**	**<0.001**
Mlig-pro34	-0.25	0.2983	0.43	0.29	0.58
Mlig-pro35	0.26	0.3025	0.42	0.50	0.78
Mlig-pro36	-0.54	0.3041	0.07	0.02	0.11
Mlig-pro37	0.22	0.2996	0.49	0.54	0.78
Mlig-pro38	-0.53	0.3040	0.08	0.01	0.07

Mlig-pro39	0.27	0.3009	0.39	0.34	0.61
Mlig-pro42	-0.12	0.2991	0.72	0.60	0.80
Mlig-pro44	-0.07	0.2996	0.83	0.76	0.84
Mlig-pro45	-0.18	0.3006	0.58	0.54	0.78
Mlig-pro46	0.11	0.3003	0.73	0.69	0.81
Mlig-pro48	-0.47	0.2985	0.13	0.03	0.13
Mlig-pro49	-0.09	0.2993	0.78	0.71	0.82
Mlig-pro51	0.19	0.3001	0.55	0.65	0.80
Mlig-pro52	-0.14	0.3015	0.67	0.61	0.80
Mlig-pro53	0.20	0.3013	0.54	0.46	0.78
Mlig-pro55	-0.37	0.2980	0.24	0.11	0.31
Mlig-pro56	-0.47	0.2986	0.13	0.05	0.17
Mlig-pro57	-0.12	0.3018	0.70	0.73	0.83
Mlig-pro58	0.01	0.2999	0.98	0.97	0.99
Mlig-pro59	0.33	0.3027	0.29	0.10	0.30
Mlig-pro60	-0.04	0.2992	0.91	0.87	0.92
Mlig-pro61	0.40	0.3001	0.20	0.29	0.58
Mlig-pro63	0.05	0.3005	0.88	0.86	0.92
Mlig-pro64	0.14	0.2990	0.67	0.66	0.80
Mlig-pro69	-0.10	0.2990	0.75	0.64	0.80
Mlig-pro70	0.11	0.3017	0.72	0.67	0.80
Mlig-pro73	0.39	0.2994	0.21	0.30	0.58
Mlig-pro74	0.14	0.3009	0.67	0.64	0.80
2216.1	-0.01	0.2975	0.97	0.96	0.99
23926	-0.07	0.3030	0.82	0.80	0.88
44523	-0.20	0.3031	0.53	0.43	0.75
471.4	-0.19	0.3002	0.56	0.49	0.78

Table S2. Related to Figure 2: Primer sequences used in the synthesis of dsRNA for RNAi knock-down of candidate seminal fluid transcripts. Underlined sequences are Sp6-promotor of Forward (*F*) and T7-promotor of Reverse (*R*) primers.

SFP Transcripts	Primer sequence
Mlig-pro5	***F***: CATTTAGGTGACACTATAGAAGTGATTATTCTCGTTGCTGCTC
	R: GGATCCTAATACGACTCACTATAGGATATTGGGTCACCGCAGTTG
Mlig-pro6	***F***: CATTTAGGTGACACTATAGAAGAATCTGTGCTCAAGCAGACTGG
	R: GGATCCTAATACGACTCACTATAGGCGGTCGCTCAAGTCGGTATC
Mlig-pro30	***F***: CATTTAGGTGACACTATAGAAGCTGTTTTAAGTCTGACGCATCG
	R: GGATCCTAATACGACTCACTATAGGCTTGCATAGGCAATCACTGG
Mlig-pro31	***F***: CATTTAGGTGACACTATAGAAGAAACGAGCAATCAACAGAACTACG
	R: GGATCCTAATACGACTCACTATAGGGGAGATCATATCTTTCCAGTCAGC
Mlig-pro32	***F***: CATTTAGGTGACACTATAGAAGAAATGCTACTGCGGATTTACG
	R: GGATCCTAATACGACTCACTATAGGGCTCATGTCGTATTTGTTTTGC
Mlig-pro33	***F***: CATTTAGGTGACACTATAGAAGATAGAAATAAGCCACCAACTCAGC
	R: GGATCCTAATACGACTCACTATAGGCATACTCTTGCACGATCACTACG

Table S3. Related to Figure 2: Descriptive statistics for the mating frequency and donor's own suck propensity of RNAi knock-down/control treatments and comparisons between knock-down treatments with control group. Welch two sample t-tests were performed to evaluate the differences between the control group and each knock-down of seminal fluid candidate transcripts for (a) their average mating frequency during 2.5 hrs period of pairing and (b) the suck propensity of the donor itself. Adjusted *P*-values derived from False Discovery Rate corrections.

(a) Mating frequency

	n	mean	sd	*t*	df	*P*	Adj. *P*
Control	30	22.97	9.13				
Mlig-pro5	29	28.10	9.35	2.14	56.80	**0.04**	0.26
Mlig-pro6	32	22.03	7.69	-0.44	56.88	0.67	0.81
Mlig-pro30	26	25.31	9.07	0.96	52.97	0.34	0.81
Mlig-pro31	27	23.59	10.08	0.25	52.77	0.81	0.81
Mlig-pro32	20	23.65	7.34	0.29	46.20	0.77	0.81
Mlig-pro33	31	22.32	8.07	-0.29	57.61	0.77	0.81
Combined	27	22.30	8.68	-0.28	54.82	0.78	0.81

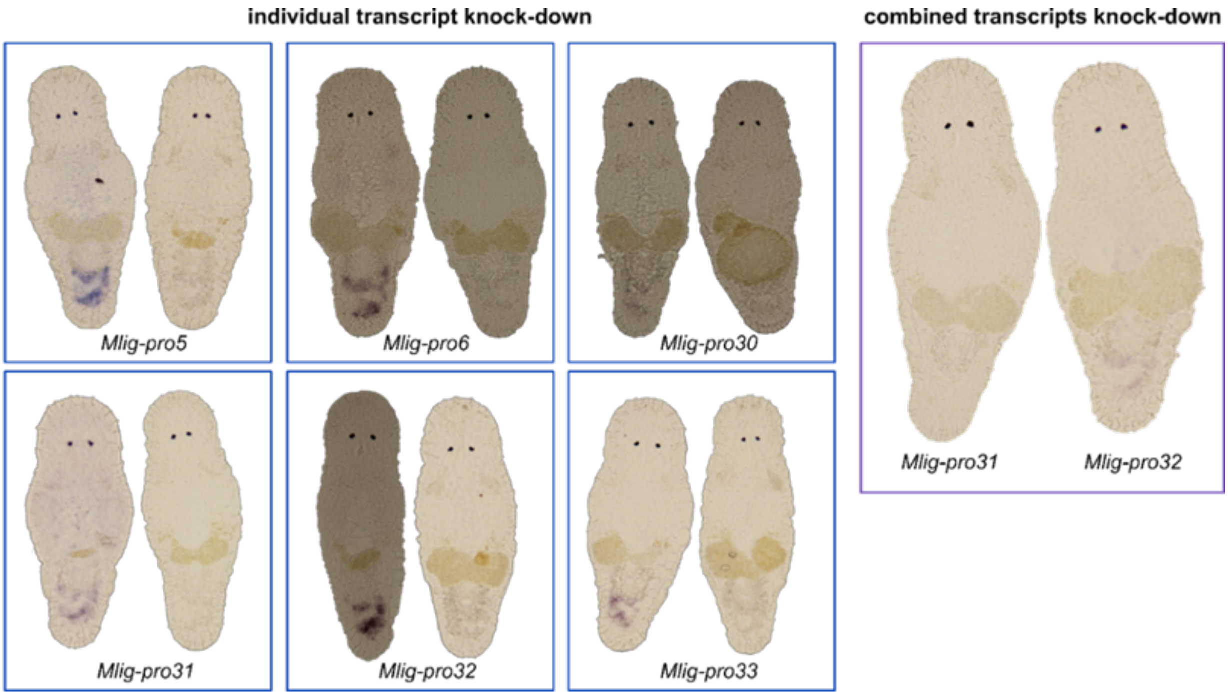
(b) Donor suck propensity

	n	mean	sd	*t*	df	*P*	Adj. *P*
Control	30	0.31	0.15				
Mlig-pro5	29	0.31	0.13	0.01	56.49	0.99	0.99
Mlig-pro6	32	0.33	0.16	0.50	59.96	0.62	0.95
Mlig-pro30	26	0.32	0.14	0.24	53.59	0.81	0.95
Mlig-pro31	27	0.35	0.15	1.08	54.52	0.28	0.95
Mlig-pro32	20	0.32	0.11	0.34	47.65	0.74	0.95
Mlig-pro33	31	0.30	0.12	-0.30	56.18	0.77	0.95
Combined	27	0.42	0.17	2.53	53.05	**0.02**	0.10

Table S4. Related to Figure 2: Descriptive statistics for defensive sperm competitive ability (P_1) scores of RNAi knock-down treatments of seminal fluid candidates and comparisons between control and knock-down treatments. Generalized linear models with a quasi-binomial error distribution and logit link function were fitted to test for the effect of each knock-down independently. The models included a response variable which is a matrix where the first column is the number of the focal worm's offspring and the second column is the number of GFP(+) offspring. *P*-value adjustments were done by performing False Discovery Rate corrections.

	n	mean	sd	Estimate	z	*P*	*Adj.*
Control	28	0.23	0.34	-1.45			
Mlig-pro5	27	0.42	0.39	0.96	1.97	**0.05**	0.21
Mlig-pro6	31	0.35	0.37	0.70	1.48	0.14	0.21
Mlig-pro30	24	0.38	0.40	0.76	1.47	0.15	0.21
Mlig-pro31	25	0.36	040	0.37	0.70	0.49	0.57
Mlig-pro32	19	0.28	0.37	0.32	0.56	0.58	0.58
Mlig-pro33	30	0.33	0.40	0.73	1.49	0.14	0.21
Combined	27	0.38	0.36	0.90	1.89	0.07	0.21

Figure S1. Related to Figure 2: Representative pictures of specimens from whole-mount *in situ* hybridization (ISH) of RNAi knock-downs. ISH was performed for at least four specimens of each control and knock-down treatment. Control samples (left-hand side of each panel) are compared to RNAi samples (right-hand side) for their ISH pattern for each single transcript knock-down. In addition, the ISH pattern of two functionally characterized seminal fluid transcripts (*Mlig-pro31* and *Mlig-pro32*) are shown for the combined knock-down treatment. In each case, note the blue-stained region in the tail of worms indicating expression of the relevant transcript in the prostate gland cells of controls, absent from RNAi knockdown worms. There was one possible exception to this pattern, with the ISH staining pattern indicating some residual *Mlig-pro32* expression in the combined knock-down treatment.



Supplemental Video S1

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